

Bayesian Deep Learning for Uncertainty Quantification

Abstract

Bayesian deep learning has become one of the central paradigms for uncertainty-aware machine learning because it combines the expressive function classes of deep neural networks with the probabilistic semantics of Bayesian inference. In the classical Bayesian formulation, uncertainty enters through a prior over latent functions or parameters and is updated into a posterior after observing data; in deep learning, this update is complicated by extreme dimensionality, multimodality, non-identifiability, and the computational burden of modern architectures. As a result, the field has developed an unusually broad spectrum of exact, approximate, and surrogate methods: Markov chain Monte Carlo and stochastic-gradient MCMC, variational inference in weight space and function space, dropout-based approximations, Laplace methods, posterior linearization, last-layer Bayes, particle-based methods, stochastic-weight averaging, and Bayesian-inspired ensembles [1–31]. In parallel, the notion of uncertainty itself has become more nuanced. Current practice distinguishes aleatoric uncertainty arising from irreducible data noise, epistemic uncertainty arising from limited knowledge about parameters or functions, and related notions such as distributional uncertainty, model misspecification, and decision uncertainty under shift [32–40].

This survey reviews Bayesian deep learning for uncertainty quantification from the perspective of posterior modeling, approximation quality, evaluation methodology, and downstream decision-making. The paper emphasizes a unifying view: most successful methods can be interpreted as approximating the posterior predictive distribution rather than merely producing a scalar confidence score. The quality of uncertainty estimation therefore depends not only on the flexibility of the underlying network but also on prior design, posterior approximation family, predictive marginalization strategy, and the mismatch between training and deployment distributions [24, 30, 31, 41–47]. This perspective explains why methods with elegant Bayesian interpretations do not automatically outperform simpler approximations or even non-Bayesian baselines, and why approximate Bayesian methods often need to be assessed under calibration, out-of-distribution, and selective-prediction criteria rather than standard test accuracy alone [49–63].

The survey organizes methods into coherent families and compares their statistical assumptions, approximation targets, computational scaling, and empirical behavior. It also covers major application areas, including computer vision, medical imaging, reinforcement learning, active learning, graphs, scientific machine learning, and neural operators [64–79, 93–97, 113–119, 151]. A recurring conclusion is that uncertainty quality is governed less by any single algorithmic label than by the alignment between inductive bias, approximation fidelity, and deployment objective. The paper closes by identifying open problems in prior specification, posterior diversity, large-model scaling, benchmark design, theory, and the integration of Bayesian and distribution-free uncertainty guarantees [98–111, 120–158].

Introduction

Deep neural networks are often deployed in settings where wrong predictions carry asymmetric and potentially severe costs. In such settings, the relevant question is not only whether a model predicts accurately on average, but whether it knows when it is likely to fail. Standard deterministic training pipelines do not provide this information reliably. Modern networks can achieve high accuracy while being sharply overconfident on corrupted, shifted, or out-of-distribution inputs, a phenomenon documented across image classification, medical imaging, language tasks, and decision-making benchmarks [49–63, 73–79]. Uncertainty quantification addresses this gap by attaching probabilistic semantics to prediction, enabling abstention, active data acquisition, risk-sensitive control, and more reliable human oversight [32–40, 64–72].

The Bayesian answer to uncertainty quantification is conceptually simple. Given a dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$, a prior $p(\theta)$, and a likelihood $p(y \mid x, \theta)$, the posterior predictive distribution is

$$p(y_{\setminus*} \mid x_{\setminus*}, \mathcal{D}) = \int p(y_{\setminus*} \mid x_{\setminus*}, \theta) p(\theta \mid \mathcal{D}) d\theta.$$

This expression captures two important sources of predictive uncertainty. The first is uncertainty that remains even if the model parameters were known exactly, often represented by the conditional variance of the likelihood and commonly called aleatoric uncertainty. The second is uncertainty induced by imperfect knowledge about the parameters or function, commonly called epistemic uncertainty. In regression, this can be written as a conditional variance decomposition,

$$\text{Var}[y_{\setminus*} \mid x_{\setminus*}, \mathcal{D}] = \mathbb{E}_{p(\theta \mid \mathcal{D})} [\text{Var}(y_{\setminus*} \mid x_{\setminus*}, \theta)] + \text{Var}_{p(\theta \mid \mathcal{D})} [\mathbb{E}(y_{\setminus*} \mid x_{\setminus*}, \theta)].$$

Deep learning makes this decomposition practically useful by offering highly expressive parameterizations, but it also makes Bayesian inference far more difficult than in classical models. The posterior over millions or billions of weights is typically diffuse, multimodal, non-Gaussian, and strongly dependent on symmetries induced by overparameterization. Consequently, the field has shifted from attempting “exact Bayes” everywhere to designing approximation families that preserve the aspects of the posterior predictive most relevant to the task at hand: calibration, coverage, ranking by uncertainty, robustness under shift, or decision quality [3, 16–31, 54, 91, 92, 144].

A second reason Bayesian deep learning remains central is that it provides a common language for multiple strands of uncertainty-aware modeling that otherwise look disconnected. Gaussian-process limits of neural networks connect wide architectures to exact Bayesian nonparametrics [41–47]. Variational inference connects stochastic optimization, regularization, and noisy gradient methods to approximate posterior learning [5–15, 20, 28, 29, 139]. Laplace methods connect curvature-aware optimization to tractable posterior surrogates around pretrained solutions [25–27, 130–133, 141]. Particle methods and diversity-promoting ensembles connect the posterior to sets of functionally distinct predictors [49, 106, 135–138, 158]. Last-layer and linearized approaches bridge feature learning with classical Bayesian linear regression [126–130]. Even methods widely treated as “non-Bayesian baselines,” such as deep ensembles or some deterministic uncertainty layers, are often best understood as approximating or emulating a posterior predictive distribution under computational constraints [24, 49, 106, 120, 135, 158].

At the same time, the literature has clarified several misconceptions. First, Bayesian semantics alone do not guarantee useful uncertainty in practice. The prior may be poorly specified, the posterior may be severely approximated, or the architecture may induce function-space behavior that is unsuitable for the deployment domain [30, 31, 148, 149, 155]. Second, standard test log-likelihood or accuracy can hide serious uncertainty failures; calibration, distribution shift, selective prediction, and coverage diagnostics are often more revealing [50–57, 144, 145]. Third, single-number distinctions between aleatoric and epistemic uncertainty can be fragile under misspecification, especially for multimodal targets, heteroscedastic settings, or out-of-distribution inputs [32–40, 121, 145]. Fourth, scalability remains a defining issue: methods that behave well on small regression benchmarks may not transfer directly to convolutional networks, transformers, language models, or neural operators [23, 29, 97, 108, 120, 127].

This survey is organized around these observations. Rather than treating Bayesian deep learning as synonymous with any one inference algorithm, it views the field through the joint lens of modeling choice and approximation target. The Methods section first introduces probabilistic foundations and prior design, then examines posterior inference families, structured approximations, evaluation strategies, and architectural or domain-specific extensions. The discussion emphasizes what each method approximates, what uncertainty signal it preserves, and where its limitations become visible. The final section distills open research problems that now define the frontier of the field: more meaningful priors, stronger function-space approximations, large-scale deployment with principled uncertainty, uncertainty under distribution shift and misspecification, and tighter connections between Bayesian, decision-theoretic, and distribution-free uncertainty quantification [30, 31, 53–63, 98–111, 120–158].

Methods

Probabilistic foundations and prior design

The starting point of Bayesian deep learning is not a particular algorithm but a modeling decision: where to place the prior and what aspects of uncertainty the predictive distribution should represent. Early Bayesian neural network work placed priors directly on weights and biases, often isotropic Gaussian, and performed posterior inference in weight space [1–4]. This remains the dominant formulation because it integrates naturally with gradient-based deep learning and standard architectures. However, modern work has shown that the prior is not a benign implementation detail. Weight-space priors determine the induced distribution over functions, interact strongly with architectural symmetries, and can profoundly affect calibration, extrapolation, and the cold-posterior phenomenon [30, 31, 48, 152–155].

A useful distinction is between three levels of prior specification. The first is unstructured weight-space priors, typically Gaussian, scale-mixture, or heavy-tailed, chosen for convenience and analytic tractability [1–3, 7, 30, 48, 152, 155]. The second is structured priors that encode sparsity, parameter correlations, or architecture-specific inductive bias. Horseshoe priors, Gaussian scale mixtures, matrix-Gaussian posteriors combined with matched priors, and heavy-tailed priors over units are examples that better reflect learned weight statistics or encourage adaptive model complexity [86, 95, 152–155]. The third is function-space prior design, where the modeler specifies desired uncertainty behavior over inputs and uses weight-space constructions only as an implementation vehicle. This view is especially natural in wide-network Gaussian-process limits and in functional variational methods [41–47, 88, 89, 137].

The relation between priors and uncertainty is subtle. A prior that is weak in weight space can still be very informative in function space because deep architectures amplify and reparameterize local perturbations.

Conversely, priors that appear innocuous can induce highly implausible function behavior far from the data manifold. This is one reason why out-of-distribution uncertainty is not an automatic consequence of “being Bayesian”; it depends on how the prior allocates probability mass to extrapolative functions [30, 31, 42–47, 148, 149, 155]. The literature on Bayesian neural network priors has therefore shifted from default Gaussians toward learned, hierarchical, correlated, and architecture-aware priors [30, 48, 152–155].

Likelihood design is equally important. In regression, Gaussian likelihoods with input-dependent variance are the default mechanism for aleatoric uncertainty, but they can fail under optimization pathologies or multimodal conditional targets [32, 39, 80, 81, 145]. In classification, the usual categorical likelihood captures label noise only indirectly, which motivates alternatives such as Dirichlet-output models, prior networks, and posterior networks for separating data ambiguity from distributional or epistemic uncertainty [35–39]. In structured prediction and scientific machine learning, the likelihood may combine observational noise, PDE residuals, latent constraints, and simulator discrepancy; here, uncertainty semantics become deeply model dependent [113–119].

One of the most important conceptual developments in the field is the growing separation between uncertainty types and uncertainty estimators. Aleatoric and epistemic uncertainty are conceptual categories, whereas predictive entropy, mutual information, variance of logits, posterior standard deviation, and interval width are estimators derived from specific models. Different estimators can behave very differently under the same conceptual label, and several surveys now caution against treating them as interchangeable [32–40, 109–111]. For example, predictive entropy is sensitive to both class overlap and model disagreement, while the mutual information between predictions and parameters is often used as a more specific measure of epistemic uncertainty in classification [40, 64–66, 124]. This distinction becomes crucial in active learning and out-of-distribution detection, where one often wants high uncertainty only when additional data could actually reduce it.

Posterior inference in Bayesian neural networks

The central computational problem is approximating $p(\theta \mid \mathcal{D})$ or, more pragmatically, the posterior predictive. The most statistically direct family is MCMC. Neal’s early work established the viability of full Bayesian inference for small neural networks via Hamiltonian or related samplers, but modern high-dimensional models require stochastic-gradient approximations [3, 16–21]. Stochastic Gradient Langevin Dynamics adds carefully scaled noise to minibatch gradients, while SGHMC and thermostat variants incorporate momentum or auxiliary dynamics to improve mixing [16–19]. Preconditioned variants adapt to local curvature, and cyclical schedules improve mode discovery in multimodal posteriors [19, 21, 140]. These methods are attractive because they target the posterior more directly than variational surrogates, but they remain expensive, sensitive to hyperparameters, and difficult to diagnose in overparameterized deep models. Their computational cost often limits their role to small and medium-scale benchmarks, posterior-quality studies, and specialized domains where higher-fidelity uncertainty justifies the expense [20, 21, 31, 140, 144].

Variational inference is the dominant scalable alternative. The basic idea is to introduce a tractable family $q_\phi(\theta)$ and optimize an evidence lower bound. Graves’ variational neural network formulation and Bayes by Backprop revived this approach for modern deep learning by combining reparameterization-based gradients with diagonal Gaussian surrogates [6, 7]. Local reparameterization and Flipout reduced gradient variance, making weight-space VI practical for minibatch training [8, 13]. Yet mean-field VI is well known to underestimate posterior dependence and often produce overconfident or poorly shaped predictive

uncertainty, especially “between” observed regions [91, 92, 144]. This problem is not only one of underdispersion; it is also geometric, because independence in weight space can imply highly constrained diversity in function space [91, 92].

Several lines of work improve expressiveness while retaining scalability. Multiplicative normalizing flows enrich the posterior family beyond mean-field assumptions [12]. Bayesian hypernetworks generate complex posterior samples through an auxiliary neural network, amortizing dependence patterns among weights [104]. Matrix-Gaussian and structured variational families preserve layerwise correlations [86]. Rank-1 factors and related low-rank updates greatly reduce the parameter overhead of Bayesianizing large models while maintaining useful posterior flexibility [108]. Natural-gradient and Gauss–Newton-inspired updates connect VI to second-order optimization, yielding algorithms such as noisy natural gradient and weight-perturbed Adam/Vadam that can be integrated with standard deep-learning pipelines [28, 29, 139].

Expectation-propagation-style objectives and generalized divergences offer another route. Probabilistic backpropagation replaces exact gradient-based VI with moment matching, originally for smaller networks [14]. Black-box α -divergence minimization generalizes the variational objective, sometimes improving predictive uncertainty by targeting regions that KL-based VI neglects [15]. These methods highlight an important lesson: the choice of divergence matters because Bayesian deep-learning posteriors are typically multimodal and broad; “reverse-KL” style approximations can collapse uncertainty precisely where uncertainty is most needed for decision-making [15, 24, 31, 91, 92].

Function-space inference emerged partly in response to these shortcomings. Instead of approximating a weight posterior and hoping it induces a good predictive process, functional variational Bayesian neural networks regularize directly in the space of stochastic processes over inputs [88]. Function-space particle optimization similarly moves particles to approximate the posterior over functions rather than over parameters [137]. These methods can align approximation quality more closely with what ultimately matters for uncertainty quantification, namely the posterior predictive over the input domain. They also connect naturally to Gaussian-process priors and wide-network limits [41–47, 88, 137]. However, they introduce their own challenges. Function-space objectives depend on the choice of measurement sets, kernel estimators, or representative input distributions, which can create blind spots off the sampled support [88, 137]. In practice, they often improve extrapolative behavior on controlled benchmarks, but scaling them to large architectures and arbitrary modalities remains challenging.

Particle-based Bayesian inference provides a middle ground between VI and MCMC. Stein variational gradient descent evolves a set of particles toward the posterior using deterministic transport combined with a repulsive kernel [138]. Applied to deep learning, this perspective clarifies why ordinary ensembles can suffer from particle collapse in function space and motivates repulsive or function-space ensemble updates [135–137]. The Stein literature also exposes the importance of the geometry in which similarity is measured: weight-space kernels can be misleading in overparameterized models, while function-space or hybrid kernels better preserve posterior diversity relevant to prediction [136, 137]. These insights have influenced both explicitly Bayesian particle approximations and more heuristic ensemble-based methods.

Deep Gaussian processes occupy a related but distinct branch of Bayesian deep learning. Rather than starting from finite-width neural-network parameters, deep GPs compose latent Gaussian processes across layers, producing a fully Bayesian hierarchy over functions [117–119]. Their uncertainty estimates are often more natural than those of weight-space BNNs, especially in low-data regimes, but inference is challenging because uncertainty must be propagated through deep latent function compositions. Variational

compression and doubly stochastic variational inference made deep GPs much more scalable, yet they remain computationally specialized relative to mainstream neural architectures [117–119]. Conceptually, however, they serve as an important counterpoint: Bayesian deep learning is broader than Bayesian neural networks, and many of the field’s design principles—layerwise uncertainty propagation, prior specification in function space, and approximate posterior factorization—are often more transparent in deep GPs than in BNNs.

Structured approximations and Bayesian surrogates

A large portion of recent progress has come from approximations that sit between full Bayesianization and deterministic training. These methods embrace the fact that, for uncertainty quantification, one often cares more about a faithful posterior predictive around a strong pretrained solution than about perfect global posterior inference.

The Laplace family exemplifies this trend. A Laplace approximation fits a Gaussian around a mode, typically a MAP solution, using local curvature information [25–27]. For deep networks, exact Hessians are infeasible, so practical implementations rely on generalized Gauss–Newton, diagonal, block-diagonal, Kronecker-factored, or last-layer approximations [25, 26]. Laplace Redux synthesized the modern recipe and showed that many common objections to Laplace methods—poor accuracy, impossibility at scale, lack of practical relevance—were overstated [26]. Subsequent work improved their predictive behavior through local linearization, mixtures of Laplace approximations, accelerated linearized Laplace using kernel/Nyström ideas, and Riemannian variants that adapt better to parameter geometry [130–133]. In practice, Laplace methods are compelling because they are post hoc, easy to attach to pretrained models, and often competitive on calibration and out-of-distribution metrics when carefully implemented [25–27, 130–133].

Linearized and last-layer Bayesian models follow the same philosophy but push the approximation further. If feature extraction is treated as deterministic and only the output layer is Bayesian, one obtains exact or near-exact Bayesian linear regression in the final representation space [126–128]. This produces closed-form predictive variances and much lower computational cost than full BNNs. The success of Bayesian last-layer methods suggests that a large portion of practically useful uncertainty can be recovered from uncertainty over task-specific readout weights, provided features are well learned. Recent work has expanded this family with latent-derivative last layers, variational Bayesian last layers, and improved marginal-likelihood-based training for extrapolation [126–128]. These methods are particularly attractive for large pretrained backbones and fine-tuning scenarios because they preserve deployment efficiency.

Subnetwork and posterior-subspace methods aim to Bayesianize only a subset of parameters deemed most important for uncertainty. Subnetwork inference formalizes this idea by selecting a low-dimensional posterior subspace, while stochastic-weight averaging and SWAG approximate posterior mass along optimization trajectories [22, 23, 129]. SWAG in particular can be interpreted as combining the geometry of wide optima with a low-rank-plus-diagonal Gaussian posterior. It often provides strong uncertainty estimates at modest additional cost, especially when full Bayesianization is prohibitive [22, 23]. These methods underscore a practical theme: high-quality uncertainty does not always require distributing all parameters. What matters is whether the retained stochastic subspace spans functionally relevant predictive variation.

Dropout-based approaches sit at the opposite end of the design spectrum: they obtain uncertainty by reusing standard regularization mechanisms with a Bayesian interpretation. MC dropout remains one of the

most widely used approximate Bayesian methods because it is simple, architecture-agnostic, and inexpensive [9, 10, 64]. Yet its limitations are now well documented. It often underestimates uncertainty between data regions, can be sensitive to dropout placement and rate, and may fail to represent rich posterior dependence [54, 91, 92, 146]. Stochastic batch normalization provides a related route by treating batch-statistics noise as posterior sampling, again with very low engineering overhead [107]. These methods remain important baselines because of their accessibility, but the field increasingly views them as useful approximations rather than adequate defaults.

Ensembles occupy an ambiguous but influential place in Bayesian deep learning. Standard deep ensembles are not exact Bayes, because independently trained MAP solutions do not sample a coherent posterior by default [24, 49]. Nevertheless, ensembles are among the strongest uncertainty baselines known and often outperform approximate BNNs under calibration and distribution shift [49, 53, 54, 146]. Recent theory and methodology have narrowed this gap. Anchored ensembles, repulsive deep ensembles, and Stein-inspired ensemble dynamics reinterpret ensembles as approximate Bayesian particle methods and explicitly encourage posterior diversity in either weight or function space [106, 135–138, 158]. This line of work is important for two reasons. Empirically, it confronts the fact that high-quality uncertainty often comes from diversity over good predictors rather than from a single convenient posterior family. Conceptually, it suggests that “Bayesian” should be understood not as a binary property but as a spectrum of posterior approximation principles.

The following table summarizes the main approximation families discussed above.

Method family	Approximation target	Main strengths	Main limitations	Representative references
SG-MCMC	Weight posterior	High-fidelity posterior samples; strong baseline for posterior studies	Expensive; mixing and tuning are difficult at scale	[16–21, 140]
Mean-field and structured VI	Weight posterior	Scalable; end-to-end training; broad architectural coverage	Can under-represent dependence and function-space diversity	[6–15, 28, 29, 86, 104, 108, 139]
Function-space VI and particle methods	Posterior over functions	Better alignment with predictive uncertainty; natural GP connections	Requires support/ input design; harder to scale	[88, 89, 137, 138]
Dropout and stochastic regularization	Implicit posterior surrogate	Very easy to implement; cheap deployment	Often underestimates epistemic uncertainty	[9, 10, 64, 107]

Method family	Approximation target	Main strengths	Main limitations	Representative references
Laplace and linearized Laplace	Local Gaussian posterior around a mode	Post hoc; strong calibration-cost trade-off	Local approximation; curvature approximation quality matters	[25–27, 130–133, 141]
Last-layer Bayes	Posterior over readout weights	Cheap; analytical predictive forms; easy to attach to large backbones	Feature uncertainty largely ignored	[126–128]
SWAG and posterior subspaces	Low-dimensional parameter subspace	Scalable; strong empirical performance	Subspace choice can miss important uncertainty directions	[22, 23, 129]
Ensembles and repulsive ensembles	Set of diverse predictors approximating posterior predictive	Empirically robust and competitive under shift	Memory/inference cost; Bayesian semantics depend on construction	[24, 49, 106, 135–138, 158]

Evaluation, robustness, and decision-making

Uncertainty quantification cannot be evaluated solely by accuracy. A model can be accurate and still unusable if it is confidently wrong at the moments when caution is required. For this reason, Bayesian deep learning has developed a rich evaluation toolbox centered on proper scoring rules, calibration, robustness to shift, coverage, and decision quality [49–57, 109–111].

For probabilistic prediction, the most principled scalar metrics are proper scoring rules such as negative log-likelihood and the Brier score, because they reward calibrated sharp distributions rather than point accuracy alone [49–53]. In practice, however, these metrics can obscure failure modes. Yao and colleagues showed that methods ranking well under log-likelihood can still produce poor uncertainty decompositions or misleading confidence behavior [144]. Calibration metrics such as expected calibration error, reliability diagrams, and classwise or adaptive calibration statistics therefore remain standard for classification, while interval coverage, calibration curves, and sharpness are common in regression [50–56]. Kuleshov and colleagues highlighted that regression uncertainty itself may need calibration post hoc, not just classification probabilities [51].

Selective prediction and abstention offer a more decision-centric perspective. If a model can defer predictions when uncertain, then the relevant object is the risk-coverage curve, not merely the average log score. SelectiveNet exemplifies end-to-end optimization for this setting, while Bayesian uncertainties often serve as abstention scores even when the base model is trained independently [123]. This perspective is especially relevant in medicine, scientific workflows, and autonomous systems, where the purpose of uncertainty is often triage rather than direct probability reporting [73–79, 113–116, 123].

Conformal prediction adds another axis by providing finite-sample coverage guarantees under exchangeability, independent of Bayesian correctness. Conformalized quantile regression is particularly important because it combines flexible deep predictive models with distribution-free interval validity [122]. From a Bayesian deep-learning perspective, conformal methods are not replacements so much as complements. Bayesian models provide adaptive, semantically rich uncertainty scores, while conformal wrappers can correct residual miscalibration and convert scores into guaranteed sets or intervals. This hybridization is likely to become increasingly important in high-stakes deployment.

Distribution shift remains the most stringent contemporary test. The benchmark of Ovadia and colleagues demonstrated that many methods that appear satisfactory in-domain degrade sharply under corruption and shift [53]. Subsequent studies in computer vision and OOD detection confirmed that deep ensembles often outperform lightweight Bayesian approximations, while posterior quality depends strongly on the architecture, metric, and shift type [54, 58–63, 146–149]. A key lesson is that uncertainty under shift is not a single problem. Covariate shift, semantic OOD, label shift, and adversarial perturbations stress different aspects of the posterior predictive. Methods such as prior networks, posterior networks, SNGP, and repulsive deep ensembles address different subsets of this space, but none solves it universally [35–37, 63, 120, 135, 158].

The table below summarizes commonly used evaluation criteria and what they actually measure.

Criterion	Typical use	What it measures well	What it can miss	Representative references
Negative log-likelihood	Classification and regression	Global probabilistic fit	Can reward overconfident but accurate in-domain behavior	[49–53, 144]
Brier score	Classification	Mean-squared probability error	Less sensitive to tail failures than log score	[49, 52]
ECE and reliability diagrams	Classification calibration	Confidence–accuracy alignment	Sensitive to binning; weak under severe shift	[50, 52, 56]
Regression calibration / interval coverage	Regression	Probability or interval validity	Coverage alone does not assess sharpness	[51, 122, 145]
AUROC/AUPR for OOD or misclassification	Detection	Ranking of uncertainty scores	Does not assess full predictive calibration	[57–63, 146–149]
Risk–coverage curves	Selective prediction	Usefulness of uncertainty for abstention	Depends on application-specific cost trade-offs	[123]

Criterion	Typical use	What it measures well	What it can miss	Representative references
Acquisition efficiency	Active learning	Whether uncertainty selects informative points	Sensitive to task noise and pool bias	[64–67, 124, 125, 156]
Regret / sample efficiency	RL, BO, bandits	Decision value of posterior uncertainty	May confound representation learning with uncertainty quality	[68–72, 93, 94, 157]

The downstream literature further shows that “good uncertainty” is task dependent. In active learning, the most useful signal is often model disagreement or mutual information rather than raw entropy, because one wants uncertainty that additional labels can reduce [64–67, 124, 125, 156]. In reinforcement learning and bandits, posterior sampling quality matters because insufficient diversity leads to under-exploration [68–72, 94]. In Bayesian optimization, local accuracy and calibrated uncertainty in unexplored regions are both required; robust BNN-based surrogates and neural-linear models attempt to satisfy this balance when Gaussian processes struggle with scale or representation complexity [93, 94, 157].

Domain-specific and modern architectural extensions

Computer vision was the earliest large-scale proving ground for Bayesian deep learning. Bayesian SegNet and related dense-prediction models demonstrated that approximate posterior sampling can provide uncertainty maps useful for semantic segmentation and scene understanding [73]. Camera relocation, medical diagnosis, and lesion segmentation further established that uncertainty can be clinically or operationally useful, not merely statistically elegant [74–79]. In vision, however, the literature also revealed a persistent cost-quality tension. Full BNNs and multi-sample methods increase latency and memory significantly, which motivates test-time augmentation, last-layer uncertainty, or ensemble surrogates in practical systems [73–79, 146].

Medical imaging occupies a distinctive subfield because the uncertainty target is often localized, structure specific, and deeply entangled with annotation noise and class imbalance. Bayesian QuickNAT, uncertainty-aware lesion segmentation, and safer neuroimage enhancement are representative examples where uncertainty is used for quality control, failure discovery, and downstream clinician interaction [76–79]. These applications also stress that voxelwise or pixelwise uncertainty maps are not automatically calibrated at a clinically meaningful scale; aggregation, calibration, and decision-specific postprocessing remain important [75–79].

Sequence models and language applications raise additional issues. Recurrent dropout and Bayesian recurrent neural networks showed that uncertainty-aware sequential modeling is possible with tractable approximations [10, 150]. Active learning in NLP highlighted that acquisition behavior can differ substantially from image tasks because sequence length, structure, and annotation granularity distort uncertainty scores [125]. At larger scales, Bayesian Layers demonstrated that uncertainty modules can be inserted into transformers and other industrial architectures, including very large models, although reliable large-language-model uncertainty remains a frontier rather than a solved problem [97, 120]. The practical

trend in language is toward partial Bayesianization—last layers, GP heads, or lightweight variational modules—rather than fully Bayesianizing every parameter of a foundation model.

Graph learning has developed its own Bayesian variants. Bayesian GCNs incorporate uncertainty over graph structure or message-passing masks, while Bayesian GNNs with adaptive connection sampling reinterpret stochastic graph regularization as approximate Bayesian inference [95, 96]. Graph Posterior Networks go a step further and perform explicit Bayesian predictive updates on attributed graphs [151]. These models are important because graph domains expose forms of uncertainty less visible in i.i.d. tabular or image data: uncertainty in edges, sparsity of labels, homophily or heterophily mismatch, and relational propagation of confidence.

Scientific machine learning has expanded Bayesian deep learning into mechanistic and infinite-dimensional settings. Bayesian neural networks for materials modeling, adversarial or Bayesian physics-informed neural networks, and approximate Bayesian neural operators illustrate three complementary directions [113–116]. First, BNNs can provide uncertainty quantification for surrogate models in engineering and materials design [113]. Second, physics-informed Bayesian models can fuse governing equations with noisy observations, often using MCMC or variational approximations over network parameters and latent physical quantities [114, 115]. Third, neural operators extend Bayesian uncertainty from finite-dimensional prediction to operator learning over functions, where one needs uncertainty on entire solution fields rather than single outputs [116]. These domains tend to benefit disproportionately from principled priors and Bayesian semantics because data are scarce, extrapolation is unavoidable, and misspecification costs are high.

Deep Gaussian processes form a neighboring family of probabilistic deep models that remain especially relevant in scientific and low-data settings [117–119]. Their compositional nonparametric structure provides a clearer uncertainty semantics than many neural-network approximations, though at the expense of architectural flexibility. In practice, the frontier is increasingly hybrid: deterministic deep feature extractors combined with Bayesian output layers, Bayesian operators over learned latent spaces, or function-space priors induced by deep architectures [41–47, 88, 116–120, 126–128].

Finally, sparsity and continual learning have become important for uncertainty-aware deployment. Horseshoe priors, Bayesian compression, structured pruning, sparse posterior concentration results, and online Laplace approximations all connect uncertainty quantification to efficient representation and lifelong adaptation [82–85, 98, 99, 141–155]. Continual-learning methods such as uncertainty-guided learning and posterior meta-replay interpret uncertainty as a memory-preserving mechanism: parameters that are known with high certainty should change less as new tasks arrive [141–143]. This is one of the clearest examples where Bayesian uncertainty serves not only prediction, but also training dynamics and model maintenance.

Challenges and Prospects

A first unresolved challenge is prior specification. The field has convincingly shown that naive isotropic Gaussian priors are often poorly aligned with learned deep-network structure, and that more realistic priors can improve calibration and sometimes mitigate cold-posterior behavior [30, 31, 48, 152–155]. Yet prior elicitation for modern architectures is still immature. We lack generally accepted principles for specifying priors over convolutional filters, attention heads, normalization parameters, residual branches, or token embeddings that correspond to credible function-space beliefs. Future work is likely to focus on priors

derived from architecture symmetries, self-supervised pretraining, domain constraints, and empirical weight statistics rather than generic default choices.

A second challenge is approximation mismatch. Much of the literature can be read as a sequence of attempts to repair mismatches between the target posterior and the tractable family used to approximate it. Mean-field VI misses important dependence; dropout-like approximations can fail between data regions; local Gaussian posteriors can ignore multiscale multimodality; ensembles can collapse functionally when diversity is not explicitly maintained [31, 54, 91, 92, 130–138, 144, 158]. This suggests that future progress may come less from inventing yet another posterior family in isolation and more from targeting the predictive function space directly, using hybrid methods that combine local curvature, particles, subspaces, and explicit diversity constraints.

A third challenge concerns out-of-distribution uncertainty and misspecification. Empirical benchmarks show that uncertainty quality degrades sharply outside the training distribution, while theoretical and GP-limit analyses indicate that common architectures may not induce function priors with the desired decay or abstention behavior [53, 58–63, 146–149]. This is a fundamental issue, not a mere implementation bug. If the prior says that far-away inputs can still be explained confidently by smooth extrapolation, then exact posterior inference will retain this bias. A key prospect, therefore, is prior and architecture co-design for uncertainty behavior: distance-aware heads, density-sensitive latent spaces, domain-constrained priors, and Bayesian components specialized to the topology of the input space [35–37, 63, 120, 148, 149].

A fourth challenge is scaling. The success of deep learning has moved the field from small regression tasks to billion-parameter foundation models, diffusion systems, and neural operators. Exact Bayesian inference is impossible at these scales, and even approximate methods must contend with distributed training, memory constraints, and real-time inference budgets [23, 29, 97, 108, 116, 127]. This reality strongly favors partial Bayesianization: last-layer Bayes, low-rank posterior factors, posterior subspaces, or post hoc Laplace approximations attached to deterministic backbones [23, 26, 108, 126–133]. The open question is how much uncertainty fidelity is lost in these reductions, and which components of a large model are most worth Bayesianizing for a given deployment objective.

A fifth challenge is evaluation realism. Much of the field still relies on small UCI-style regressions, synthetic one-dimensional extrapolation plots, or narrow OOD benchmarks. These are useful for diagnosis, but insufficient for ranking methods intended for real decision systems [53, 55, 109–111, 146]. Future benchmarks need richer shift taxonomies, labeled abstention costs, calibration under multimodal targets, interactive data-acquisition loops, and continual or streaming updates. They also need separation between epistemic and aleatoric failure modes at the task level, not only as abstract concepts. Uncertainty is ultimately a decision interface, and it must be benchmarked accordingly.

A sixth challenge is theory. Bayesian deep learning now has partial theoretical foundations in PAC-Bayes, sparse posterior concentration, Gaussian-process limits, and some uncertainty-coverage results [41–47, 98–103]. However, there is still no comprehensive theory explaining when approximate Bayesian uncertainty remains decision-relevant under overparameterization, representation learning, and dataset shift. The most promising path may be a synthesis of viewpoints: PAC-Bayesian control of posterior complexity, function-space analyses of predictive behavior, and distribution-free wrappers such as conformalization for downstream guarantees [100–103, 122, 123]. Rather than seeking universal guarantees from one paradigm, future work may combine Bayesian adaptivity with finite-sample validity.

A final prospect lies in hybrid systems. Bayesian and non-Bayesian uncertainty methods are increasingly converging. Prior networks, posterior networks, distance-aware Gaussian-process heads, conformalized deep regressors, and ensemble-particle hybrids borrow from Bayesian reasoning without insisting on exact posterior semantics [35–37, 120–123, 135–138, 158]. The likely next phase of the field is therefore pluralistic. In many applications the best system may not be a pure BNN, but a carefully composed pipeline: a deterministic or pretrained backbone, a Bayesian last layer or local Laplace posterior, a conformal calibration stage, and a downstream abstention or acquisition rule matched to the operational objective. Bayesian deep learning will remain central in this ecosystem because it provides the language for what the uncertainty estimates are supposed to mean, even when practical systems use approximations and hybrids.

Conclusion

Bayesian deep learning for uncertainty quantification is best understood as a family of methods for approximating posterior predictive distributions in highly expressive models. Its central promise is not merely to attach error bars to neural-network outputs, but to represent uncertainty in a way that supports calibration, abstention, exploration, robustness, and scientific reasoning under limited data. The literature now makes clear that this promise depends jointly on prior design, posterior approximation, predictive marginalization, and evaluation protocol. Strong uncertainty estimates can emerge from several routes—SG-MCMC, expressive VI, Laplace methods, last-layer Bayes, posterior subspaces, or diversity-preserving ensembles—but none is uniformly dominant across scales and tasks [16–31, 49–63, 126–138].

The most important conceptual shift in the field has been away from treating uncertainty as a by-product of a Bayesian label and toward treating it as an object that must be modeled, approximated, and validated in its own right. This shift has produced better priors, richer function-space methods, stronger post hoc approximations, and more realistic benchmarks. It has also revealed persistent limits: approximate posteriors can distort uncertainty severely, exact Bayes can still be poorly aligned with desired OOD behavior, and scaling principled uncertainty to foundation-model regimes remains unfinished. These are not signs of failure. They are signs that Bayesian deep learning has matured from a collection of appealing heuristics into a research area with a clearer understanding of what uncertainty quantification requires, where current tools succeed, and where new theory and algorithms are still needed [30, 31, 53–63, 98–111, 148–158].

References

- [1] D. J. C. MacKay. Bayesian interpolation. *Neural Computation*, 4(3):415–447, 1992.
- [2] D. J. C. MacKay. A practical Bayesian framework for backpropagation networks. *Neural Computation*, 4(3):448–472, 1992.
- [3] R. M. Neal. *Bayesian Learning for Neural Networks*. Springer, 1996.
- [4] G. E. Hinton and D. van Camp. Keeping the neural networks simple by minimizing the description length of the weights. In *COLT*, 1993.
- [5] M. I. Jordan, Z. Ghahramani, T. S. Jaakkola, and L. K. Saul. An introduction to variational methods for graphical models. *Machine Learning*, 37:183–233, 1999.

- [6] A. Graves. Practical variational inference for neural networks. In *NeurIPS*, 2011.
- [7] C. Blundell, J. Cornebise, K. Kavukcuoglu, and D. Wierstra. Weight uncertainty in neural networks. In *ICML*, 2015. arXiv:1505.05424.
- [8] D. P. Kingma, T. Salimans, and M. Welling. Variational dropout and the local reparameterization trick. In *NeurIPS*, 2015. arXiv:1506.02557.
- [9] Y. Gal and Z. Ghahramani. Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. In *ICML*, 2016. arXiv:1506.02142.
- [10] Y. Gal and Z. Ghahramani. A theoretically grounded application of dropout in recurrent neural networks. In *NeurIPS*, 2016. arXiv:1512.05287.
- [11] D. Molchanov, A. Ashukha, and D. Vetrov. Variational dropout sparsifies deep neural networks. In *ICML*, 2017. arXiv:1701.05369.
- [12] C. Louizos and M. Welling. Multiplicative normalizing flows for variational Bayesian neural networks. In *ICML*, 2017. arXiv:1703.01961.
- [13] Y. Wen, P. Vicol, J. Ba, D. Tran, and R. Grosse. Flipout: Efficient pseudo-independent weight perturbations on mini-batches. In *ICLR*, 2018. arXiv:1803.04386.
- [14] J. M. Hernández-Lobato and R. Adams. Probabilistic backpropagation for scalable learning of Bayesian neural networks. In *ICML*, 2015.
- [15] J. M. Hernández-Lobato, Y. Li, M. Rowland, T. Bui, D. Hernández-Lobato, and R. Turner. Black-box α -divergence minimization. In *ICML*, 2016. arXiv:1511.03243.
- [16] M. Welling and Y. W. Teh. Bayesian learning via stochastic gradient Langevin dynamics. In *ICML*, 2011.
- [17] T. Chen, E. Fox, and C. Guestrin. Stochastic gradient Hamiltonian Monte Carlo. In *ICML*, 2014.
- [18] N. Ding, Y. Fang, R. Babbush, C. Chen, R. D. Skeel, and H. Neven. Bayesian sampling using stochastic gradient thermostats. In *NeurIPS*, 2014.
- [19] C. Li, C. Chen, D. Carlson, and L. Carin. Preconditioned stochastic gradient Langevin dynamics for deep neural networks. In *AAAI*, 2016. arXiv:1512.07666.
- [20] S. Mandt, M. D. Hoffman, and D. M. Blei. Stochastic gradient descent as approximate Bayesian inference. *Journal of Machine Learning Research*, 18(134):1–35, 2017. arXiv:1704.04289.
- [21] Y.-A. Ma, T. Chen, and E. Fox. A complete recipe for stochastic gradient MCMC. In *NeurIPS*, 2015. arXiv:1506.04696.

- [22] P. Izmailov, D. Podoprikin, T. Garipov, D. Vetrov, and A. G. Wilson. Averaging weights leads to wider optima and better generalization. In *UAI*, 2018. arXiv:1803.05407.
- [23] W. Maddox, T. Garipov, P. Izmailov, D. Vetrov, and A. G. Wilson. A simple baseline for Bayesian uncertainty in deep learning. In *NeurIPS*, 2019. arXiv:1902.02476.
- [24] A. G. Wilson and P. Izmailov. Bayesian deep learning and a probabilistic perspective of generalization. In *NeurIPS*, 2020. arXiv:2002.08791.
- [25] H. Ritter, A. Botev, and D. Barber. A scalable Laplace approximation for neural networks. In *ICLR*, 2018. arXiv:1805.10972.
- [26] E. Daxberger, A. Kristiadi, A. Immer, R. Eschenhagen, M. Bauer, and P. Hennig. Laplace Redux—Effortless Bayesian deep learning. In *NeurIPS*, 2021. arXiv:2106.14806.
- [27] A. Kristiadi, M. Hein, and P. Hennig. Being Bayesian, even just a bit, fixes overconfidence in ReLU networks. In *ICML*, 2020. arXiv:2002.10118.
- [28] M. E. Khan, D. Nielsen, V. Tangkaratt, W. Lin, Y. Gal, and A. Srivastava. Fast and scalable Bayesian deep learning by weight-perturbation in Adam. In *ICML*, 2018. arXiv:1806.04854.
- [29] K. Osawa, S. Swaroop, A. Jain, R. Eschenhagen, R. Turner, R. Yokota, and M. E. Khan. Practical deep learning with Bayesian principles. In *NeurIPS*, 2019. arXiv:1906.02506.
- [30] V. Fortuin. Priors in Bayesian deep learning: A review. *International Statistical Review*, 90(3):563–591, 2022. arXiv:2105.06868.
- [31] F. Wenzel, K. Roth, B. S. Veeling, J. Świątkowski, L. Tran, S. Mandt, J. M. Hernández-Lobato, R. Jenatton, and S. Nowozin. How good is the Bayes posterior in deep neural networks really? In *ICML*, 2020. arXiv:2002.02405.
- [32] A. Kendall and Y. Gal. What uncertainties do we need in Bayesian deep learning for computer vision? In *NeurIPS*, 2017. arXiv:1703.04977.
- [33] A. Der Kiureghian and O. Ditlevsen. Aleatory or epistemic? Does it matter? *Structural Safety*, 31(2):105–112, 2009.
- [34] E. Hüllermeier and W. Waegeman. Aleatoric and epistemic uncertainty in machine learning: An introduction to concepts and methods. *Machine Learning*, 110:457–506, 2021.
- [35] A. Malinin and M. Gales. Predictive uncertainty estimation via prior networks. In *NeurIPS*, 2018. arXiv:1802.10501.
- [36] B. Charpentier, D. Zügner, and S. Günnemann. Posterior network: Uncertainty estimation without OOD samples via density-based pseudo-counts. In *NeurIPS*, 2020. arXiv:2006.09239.

- [37] B. Charpentier, O. Borchert, D. Zügner, S. Geisler, and S. Günnemann. Natural posterior network: Deep Bayesian uncertainty for exponential family distributions. In *ICLR*, 2022. arXiv:2105.04471.
- [38] M. Sensoy, L. Kaplan, and M. Kandemir. Evidential deep learning to quantify classification uncertainty. In *NeurIPS*, 2018.
- [39] A. Amini, W. Schwarting, A. Soleimany, and D. Rus. Deep evidential regression. In *NeurIPS*, 2020. arXiv:1910.02600.
- [40] J. Depeweg, J.-M. Hernández-Lobato, F. Doshi-Velez, and S. Udluft. Decomposition of uncertainty in Bayesian deep learning for efficient and risk-sensitive learning. In *ICML*, 2018. arXiv:1710.07283.
- [41] C. K. I. Williams. Computing with infinite networks. In *NeurIPS*, 1997.
- [42] J. Lee, Y. Bahri, R. Novak, S. S. Schoenholz, J. Pennington, and J. Sohl-Dickstein. Deep neural networks as Gaussian processes. In *ICLR*, 2018. arXiv:1711.00165.
- [43] A. G. de G. Matthews, J. Hron, M. Rowland, R. Turner, and Z. Ghahramani. Gaussian process behaviour in wide deep neural networks. In *ICLR*, 2018. arXiv:1804.11271.
- [44] A. Garriga-Alonso, L. Aitchison, and L. Rasmussen. Deep convolutional networks as shallow Gaussian processes. In *ICLR*, 2019. arXiv:1808.05587.
- [45] R. Novak, L. Xiao, J. Lee, Y. Bahri, G. Yang, G. Hron, J. Abolafia, J. Pennington, and J. Sohl-Dickstein. Bayesian deep convolutional networks with many channels are Gaussian processes. In *ICLR*, 2019. arXiv:1810.05148.
- [46] A. Jacot, F. Gabriel, and C. Hongler. Neural tangent kernel: Convergence and generalization in neural networks. In *NeurIPS*, 2018. arXiv:1806.07572.
- [47] G. Yang. Scaling limits of wide neural networks with weight sharing: Gaussian process behavior, gradient independence, and neural tangent kernel derivation. In *ICLR*, 2020. arXiv:1902.04760.
- [48] M. Vladimirova, J. Verbeek, P. Mesejo, and J. Arbel. Understanding priors in Bayesian neural networks at the unit level. In *ICML*, 2019. arXiv:1810.05110.
- [49] B. Lakshminarayanan, A. Pritzel, and C. Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *NeurIPS*, 2017. arXiv:1612.01474.
- [50] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger. On calibration of modern neural networks. In *ICML*, 2017. arXiv:1706.04599.
- [51] V. Kuleshov, N. Fenner, and S. Ermon. Accurate uncertainties for deep learning using calibrated regression. In *ICML*, 2018. arXiv:1807.00263.

- [52] J. Nixon, M. W. Dusenberry, L. Zhang, G. Jerfel, and D. Tran. Measuring calibration in deep learning. In *CVPR Workshops*, 2019. arXiv:1904.01685.
- [53] Y. Ovadia, E. Fertig, J. Ren, Z. Nado, D. Sculley, S. Nowozin, J. V. Dillon, B. Lakshminarayanan, and J. Snoek. Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift. In *NeurIPS*, 2019. arXiv:1906.02530.
- [54] A. Ashukha, A. Lyzhov, D. Molchanov, and D. Vetrov. Pitfalls of in-domain uncertainty estimation and ensembling in deep learning. In *ICLR*, 2020. arXiv:2002.06470.
- [55] Z. Nado, N. Band, M. Collier, J. et al. Uncertainty baselines: Benchmarks for uncertainty and robustness in deep learning. arXiv preprint arXiv:2106.04015, 2021.
- [56] M. Minderer, J. Djolonga, R. Romijnders, et al. Revisiting the calibration of modern neural networks. In *NeurIPS*, 2021. arXiv:2106.07998.
- [57] H. Jiang, B. Kim, M. Guan, and M. Gupta. To trust or not to trust a classifier. In *NeurIPS*, 2018. arXiv:1805.11783.
- [58] D. Hendrycks and K. Gimpel. A baseline for detecting misclassified and out-of-distribution examples in neural networks. In *ICLR*, 2017. arXiv:1610.02136.
- [59] S. Liang, Y. Li, and R. Srikant. Enhancing the reliability of out-of-distribution image detection in neural networks. In *ICLR*, 2018. arXiv:1706.02690.
- [60] K. Lee, K. Lee, H. Lee, and J. Shin. A simple unified framework for detecting out-of-distribution samples and adversarial attacks. In *NeurIPS*, 2018. arXiv:1807.03888.
- [61] E. Nalisnick, A. Matsukawa, Y. W. Teh, D. Gorur, and B. Lakshminarayanan. Do deep generative models know what they don't know? In *ICLR*, 2019. arXiv:1810.09136.
- [62] D. Hendrycks, M. Mazeika, and T. Dietterich. Deep anomaly detection with outlier exposure. In *ICLR*, 2019. arXiv:1812.04606.
- [63] J. van Amersfoort, L. Smith, A. Jesson, O. Key, and Y. Gal. Uncertainty estimation using prior networks. In *NeurIPS*, 2020. arXiv:2011.11106.
- [64] Y. Gal, R. Islam, and Z. Ghahramani. Deep Bayesian active learning with image data. In *ICML*, 2017. arXiv:1703.02910.
- [65] W. H. Beluch, T. Genewein, A. Nürnberger, and J. M. Köhler. The power of ensembles for active learning in image classification. In *CVPR*, 2018.
- [66] A. Kirsch, J. van Amersfoort, and Y. Gal. BatchBALD: Efficient and diverse batch acquisition for deep Bayesian active learning. In *NeurIPS*, 2019. arXiv:1906.08158.

- [67] B. Settles. Active learning literature survey. University of Wisconsin-Madison Technical Report, 2010.
- [68] I. Osband, C. Blundell, A. Pritzel, and B. Van Roy. Deep exploration via bootstrapped DQN. In *NeurIPS*, 2016. arXiv:1602.04621.
- [69] I. Osband, J. Aslanides, and A. Cassirer. Randomized prior functions for deep reinforcement learning. In *NeurIPS*, 2018. arXiv:1806.03335.
- [70] J. Depeweg, K. U. Jonschkowski, R. Hafner, and S. Udluft. Learning and policy search in stochastic dynamical systems with Bayesian neural networks. In *ICLR*, 2017. arXiv:1605.07127.
- [71] K. Chua, R. Calandra, R. McAllister, and S. Levine. Deep reinforcement learning in a handful of trials using probabilistic dynamics models. In *NeurIPS*, 2018. arXiv:1805.12114.
- [72] K. Ciosek, M. Rowland, A. S. M. B. Wilson, and M. Ghavamzadeh. Better exploration with optimistic neural networks. In *NeurIPS*, 2019. arXiv:1910.12807.
- [73] A. Kendall, V. Badrinarayanan, and R. Cipolla. Bayesian SegNet: Model uncertainty in deep convolutional encoder-decoder architectures for scene understanding. arXiv preprint arXiv:1511.02680, 2015.
- [74] A. Kendall and R. Cipolla. Modelling uncertainty in deep learning for camera relocalization. In *ICRA*, 2016. arXiv:1509.05909.
- [75] C. Leibig, V. Allken, M. Berens, U. Wahl, S. Lang, and H. Bulk. Leveraging uncertainty information from deep neural networks for disease detection. *Scientific Reports*, 7:17816, 2017. arXiv:1703.04977.
- [76] T. Nair, D. T. Precup, D. Arnold, and T. Arbel. Exploring uncertainty measures in deep networks for multiple sclerosis lesion detection and segmentation. *Medical Image Analysis*, 59:101557, 2020. arXiv:1808.01200.
- [77] A. G. Roy, S. Conjeti, N. Navab, and C. Wachinger. Bayesian QuickNAT: Model uncertainty in deep whole-brain segmentation for structure-wise quality control. *NeuroImage*, 195:11–22, 2019. arXiv:1811.09880.
- [78] R. Tanno, D. Worrall, E. Kaden, et al. Uncertainty modelling in deep learning for safer neuroimage enhancement: Demonstration in diffusion MRI. *NeuroImage*, 225:117366, 2021. arXiv:1907.13418.
- [79] G. Wang, W. Li, S. Ourselin, and T. Vercauteren. Aleatoric uncertainty estimation with test-time augmentation for medical image segmentation with convolutional neural networks. *Neurocomputing*, 338:34–45, 2019. arXiv:1811.01839.
- [80] D. A. Nix and A. S. Weigend. Estimating the mean and variance of the target probability distribution. In *ICNN*, 1994.
- [81] C. M. Bishop. Mixture density networks. Technical Report NCRG/94/004, Aston University, 1994.

- [82] C. Louizos, K. Ullrich, and M. Welling. Bayesian compression for deep learning. In *NeurIPS*, 2017. arXiv:1705.08665.
- [83] K. Neklyudov, D. Molchanov, A. Ashukha, and D. Vetrov. Structured Bayesian pruning via log-normal multiplicative noise. In *NeurIPS*, 2017. arXiv:1705.07283.
- [84] K. Ullrich, E. Meeds, and M. Welling. Soft weight-sharing for neural network compression. In *ICLR*, 2017. arXiv:1702.04008.
- [85] B. L. Trippe and R. E. Turner. Overpruning in variational Bayesian neural networks. In *ICML Workshop on Theoretical Foundations and Applications of Deep Generative Models*, 2018. arXiv:1801.06230.
- [86] C. Louizos and M. Welling. Structured and efficient variational deep learning with matrix Gaussian posteriors. In *ICML*, 2016. arXiv:1603.04733.
- [87] N. Pawłowski, A. Brock, M. C. Lee, et al. Implicit weight uncertainty in neural networks. In *Bayesian Deep Learning Workshop, NeurIPS*, 2018. arXiv:1801.09885.
- [88] S. Sun, G. Zhang, J. Shi, and R. Grosse. Functional variational Bayesian neural networks. In *ICLR*, 2019. arXiv:1903.05779.
- [89] C. Louizos, X. Shi, K. Schutte, and M. Welling. The functional neural process. In *NeurIPS*, 2019. arXiv:1906.08324.
- [90] D. Hafner, D. Tran, A. Irpan, T. Lillicrap, and J. Davidson. Reliable uncertainty estimates in deep neural networks using noise contrastive priors. In *Bayesian Deep Learning Workshop, NeurIPS*, 2018. arXiv:1807.09289.
- [91] A. Y. K. Foong, Y. Li, J. M. Hernández-Lobato, and R. E. Turner. ‘In-between’ uncertainty in Bayesian neural networks. In *ICML Workshop on UDL*, 2019. arXiv:1906.11537.
- [92] A. Y. K. Foong, D. R. Burt, Y. Li, and R. E. Turner. On the expressiveness of approximate inference in Bayesian neural networks. In *NeurIPS*, 2020. arXiv:1909.00719.
- [93] J. Snoek, O. Rippel, K. Swersky, et al. Scalable Bayesian optimization using deep neural networks. In *ICML*, 2015. arXiv:1502.05700.
- [94] C. Riquelme, G. Tucker, and J. Snoek. Deep Bayesian bandits showdown: An empirical comparison of Bayesian deep networks for Thompson sampling. In *ICLR*, 2018. arXiv:1802.09127.
- [95] Y. Zhang, S. Pal, M. Coates, and D. Üstebay. Bayesian graph convolutional neural networks for semi-supervised classification. In *AAAI*, 2019. arXiv:1811.11103.
- [96] A. Hasanzadeh, E. Hajiramezanali, S. Boluki, et al. Bayesian graph neural networks with adaptive connection sampling. In *ICML*, 2020. arXiv:2006.04064.

- [97] D. Tran, M. W. Dusenberry, M. van der Wilk, and D. Hafner. Bayesian layers: A module for neural network uncertainty. In *NeurIPS*, 2019. arXiv:1812.03973.
- [98] N. G. Polson and V. Ročková. Posterior concentration for sparse deep learning. In *NeurIPS*, 2018. arXiv:1809.09001.
- [99] Y. Wang and V. Ročková. Uncertainty quantification for sparse deep learning. In *AISTATS*, 2020. arXiv:1910.14654.
- [100] D. McAllester. PAC-Bayesian model averaging. In *COLT*, 1999.
- [101] M. Seeger. PAC-Bayesian generalisation error bounds for Gaussian process classification. *Journal of Machine Learning Research*, 3:233–269, 2002.
- [102] G. K. Dziugaite and D. M. Roy. Computing nonvacuous generalization bounds for deep (stochastic) neural networks with many more parameters than training data. In *UAI*, 2017. arXiv:1703.11008.
- [103] B. Neyshabur, S. Bhojanapalli, D. McAllester, and N. Srebro. A PAC-Bayesian approach to spectrally-normalized margin bounds for neural networks. In *ICLR*, 2018. arXiv:1707.09564.
- [104] D. Krueger, C.-W. Huang, R. Islam, R. Turner, A. Lacoste, and A. Courville. Bayesian hypernetworks. In *Bayesian Deep Learning Workshop, NeurIPS*, 2017. arXiv:1710.04759.
- [105] J. Antorán, J. U. Allingham, and J. M. Hernández-Lobato. Depth uncertainty in neural networks. In *NeurIPS*, 2020. arXiv:2006.08437.
- [106] T. Pearce, F. Leibfried, A. Brintrup, M. Zaki, and A. Neely. Uncertainty in neural networks: Approximately Bayesian ensembling. In *AISTATS*, 2020. arXiv:1810.05546.
- [107] A. Atanov, A. Ashukha, D. Molchanov, K. Neklyudov, and D. Vetrov. Uncertainty estimation via stochastic batch normalization. In *ISNN*, 2019. arXiv:1802.04893.
- [108] M. W. Dusenberry, G. Jerfel, Y. Wen, Y.-A. Ma, J. Snoek, K. Heller, B. Lakshminarayanan, and D. Tran. Efficient and scalable Bayesian neural nets with rank-1 factors. In *ICML*, 2020. arXiv:2005.07186.
- [109] M. Abdar, F. Pourpanah, S. Hussain, et al. A review of uncertainty quantification in deep learning: Techniques, applications and challenges. *Information Fusion*, 76:243–297, 2021. arXiv:2011.06225.
- [110] J. Gawlikowski, C. R. N. Tassi, M. Ali, et al. A survey of uncertainty in deep neural networks. *Artificial Intelligence Review*, 56:1513–1589, 2023. arXiv:2107.03342.
- [111] W. He, Z. Jiang, T. Xiao, Z. Xu, and Y. Li. A survey on uncertainty quantification methods for deep learning. *ACM Computing Surveys*, 58(7), 2025. arXiv:2302.13425.

- [112] L. V. Jospin, H. Laga, F. Boussaid, W. Buntine, and M. Bennamoun. Hands-on Bayesian neural networks —A tutorial for deep learning users. *IEEE Computational Intelligence Magazine*, 17(2):29–48, 2022. arXiv:2007.06823.
- [113] A. Olivier, M. D. Shields, and L. Graham-Brady. Bayesian neural networks for uncertainty quantification in data-driven materials modeling. *Computer Methods in Applied Mechanics and Engineering*, 386:114079, 2021.
- [114] Y. Yang and P. Perdikaris. Adversarial uncertainty quantification in physics-informed neural networks. *Journal of Computational Physics*, 394:136–152, 2019.
- [115] L. Yang, X. Meng, and G. E. Karniadakis. B-PINNs: Bayesian physics-informed neural networks for forward and inverse PDE problems with noisy data. *Journal of Computational Physics*, 425:109913, 2021. arXiv:2003.06097.
- [116] E. Magnani, F. H. Case, J. A. Morales, and H. Antil. Approximate Bayesian neural operators: Uncertainty quantification for infinite-dimensional problems. In *NeurIPS Workshop on Gaussian Processes, Spatiotemporal Modeling, and Decision-making Systems*, 2022. arXiv:2208.01565.
- [117] A. C. Damianou and N. D. Lawrence. Deep Gaussian processes. In *AISTATS*, 2013. arXiv:1211.0358.
- [118] J. Hensman and N. D. Lawrence. Nested variational compression in deep Gaussian processes. arXiv preprint arXiv:1412.1370, 2014.
- [119] H. Salimbeni and M. Deisenroth. Doubly stochastic variational inference for deep Gaussian processes. In *NeurIPS*, 2017. arXiv:1705.08933.
- [120] J. Z. Liu, Z. Lin, S. Padhy, D. Tran, T. Bedrax-Weiss, and B. Lakshminarayanan. Simple and principled uncertainty estimation with deterministic deep learning via distance awareness. In *NeurIPS*, 2020. arXiv:2006.10108.
- [121] N. Tagasovska and D. Lopez-Paz. Single-model uncertainties for deep learning. In *NeurIPS*, 2019. arXiv:1811.00908.
- [122] Y. Romano, E. Patterson, and E. J. Candès. Conformalized quantile regression. In *NeurIPS*, 2019. arXiv:1905.03222.
- [123] Y. Geifman and R. El-Yaniv. SelectiveNet: A deep neural network with an integrated reject option. In *ICML*, 2019. arXiv:1901.09192.
- [124] N. Houlsby, F. Huszár, Z. Ghahramani, and M. Lengyel. Bayesian active learning for classification and preference learning. arXiv preprint arXiv:1112.5745, 2011.
- [125] A. Siddhant and Z. C. Lipton. Deep Bayesian active learning for natural language processing: Results of a large-scale empirical study. In *EMNLP*, 2018. arXiv:1808.05697.

- [126] J. Watson, J. A. Lin, P. Klink, J. Pajarinen, and J. Peters. Latent derivative Bayesian last layer networks. In *AISTATS*, 2021.
- [127] J. Harrison, J. Willes, and J. Snoek. Variational Bayesian last layers. In *ICLR*, 2024. arXiv:2404.11599.
- [128] F. Fiedler and S. Lucia. Improved uncertainty quantification for neural networks with Bayesian last layer. *IEEE Access*, 11:125845–125858, 2023. arXiv:2302.10975.
- [129] E. Daxberger, E. Nalisnick, J. U. Allingham, J. Antorán, and J. M. Hernández-Lobato. Expressive yet tractable Bayesian deep learning via subnetwork inference. arXiv preprint arXiv:2010.14689, 2020.
- [130] A. Immer, M. Korzepa, and M. Bauer. Improving predictions of Bayesian neural nets via local linearization. In *AISTATS*, 2021. arXiv:2008.08400.
- [131] R. Eschenhagen, E. Daxberger, P. Hennig, and A. Kristiadi. Mixtures of Laplace approximations for improved post-hoc uncertainty in deep learning. In *Bayesian Deep Learning Workshop, NeurIPS*, 2021. arXiv:2111.03577.
- [132] Z. Deng, F. Zhou, and J. Zhu. Accelerated linearized Laplace approximation for Bayesian deep learning. In *NeurIPS*, 2022. arXiv:2210.12642.
- [133] F. Bergamin, P. Rector-Brooks, B. D. Mishra, A. N. Bhatt, and S. Mandt. Riemannian Laplace approximations for Bayesian neural networks. In *NeurIPS*, 2023. arXiv:2303.08875.
- [134] S. Farquhar, M. A. Osborne, and Y. Gal. Radial Bayesian neural networks: Beyond discrete support in large-scale Bayesian deep learning. In *AISTATS*, 2020. arXiv:1907.00865.
- [135] F. D’Angelo and V. Fortuin. Repulsive deep ensembles are Bayesian. In *NeurIPS*, 2021. arXiv:2106.11642.
- [136] F. D’Angelo, V. Fortuin, and F. Wenzel. On Stein variational neural network ensembles. In *UDL Workshop, ICML*, 2021. arXiv:2106.10760.
- [137] Z. Wang, H. Li, U. Backes, and A. Krause. Function-space particle optimization for Bayesian neural networks. In *ICLR*, 2019. arXiv:1902.09754.
- [138] Q. Liu and D. Wang. Stein variational gradient descent: A general purpose Bayesian inference algorithm. In *NeurIPS*, 2016. arXiv:1608.04471.
- [139] G. Zhang, S. Sun, D. Duvenaud, and R. Grosse. Noisy natural gradient as variational inference. In *ICML*, 2018. arXiv:1712.02390.
- [140] R. Zhang, C. Li, J. Zhang, C. Chen, and A. G. Wilson. Cyclical stochastic gradient MCMC for Bayesian deep learning. In *ICLR*, 2020. arXiv:1902.03932.

- [141] H. Ritter, A. Botev, and D. Barber. Online structured Laplace approximations for overcoming catastrophic forgetting. In *NeurIPS*, 2018. arXiv:1805.07810.
- [142] S. Ebrahimi, M. Elhoseiny, T. Darrell, and M. Rohrbach. Uncertainty-guided continual learning with Bayesian neural networks. In *ICLR*, 2020. arXiv:1906.02425.
- [143] C. Henning, M. R. Cervera, F. D’Angelo, J. von Oswald, R. Traber, B. Ehret, S. Kobayashi, B. F. Grewe, and J. Sacramento. Posterior meta-replay for continual learning. In *NeurIPS*, 2021. arXiv:2109.09735.
- [144] J. Yao, W. Pan, S. Ghosh, and F. Doshi-Velez. Quality of uncertainty quantification for Bayesian neural network inference. In *ICML Workshop on UDL*, 2019. arXiv:1906.09686.
- [145] M. Seitzer, A. Tavakoli, D. Antic, and G. Martius. On the pitfalls of heteroscedastic uncertainty estimation with probabilistic neural networks. In *ICLR*, 2022. arXiv:2203.09168.
- [146] F. K. Gustafsson, M. Danelljan, and T. B. Schön. Evaluating scalable Bayesian deep learning methods for robust computer vision. In *CVPR Workshops*, 2020. arXiv:1906.01620.
- [147] J. Mitros, A. Pakrashi, and B. Mac Namee. Ramifications of approximate posterior inference for Bayesian deep learning in adversarial and out-of-distribution settings. In *ECCV Workshops*, 2020. arXiv:2009.01798.
- [148] C. Henning, F. D’Angelo, and B. F. Grewe. Are Bayesian neural networks intrinsically good at out-of-distribution detection? In *UDL Workshop, ICML*, 2021. arXiv:2107.12248.
- [149] F. D’Angelo and C. Henning. On out-of-distribution detection with Bayesian neural networks. arXiv preprint arXiv:2110.06020, 2021.
- [150] M. Fortunato, C. Blundell, and O. Vinyals. Bayesian recurrent neural networks. arXiv preprint arXiv:1704.02798, 2017.
- [151] M. Stadler, B. Charpentier, S. Geisler, D. Zügner, and S. Günnemann. Graph posterior network: Bayesian predictive uncertainty for node classification. In *NeurIPS*, 2021. arXiv:2110.14012.
- [152] T. Cui, A. Havulinna, P. Marttinen, and S. Kaski. Informative Gaussian scale mixture priors for Bayesian neural networks. arXiv preprint arXiv:2002.10243, 2020.
- [153] S. Ghosh and F. Doshi-Velez. Model selection in Bayesian neural networks via horseshoe priors. *Journal of Machine Learning Research*, 20(182):1–46, 2019. arXiv:1705.10388.
- [154] S. Ghosh, J. Yao, and F. Doshi-Velez. Structured variational learning of Bayesian neural networks with horseshoe priors. In *ICML*, 2018. arXiv:1806.05975.
- [155] V. Fortuin, A. Garriga-Alonso, S. W. Ober, F. Wenzel, G. Rätsch, R. E. Turner, M. van der Wilk, and L. Aitchison. Bayesian neural network priors revisited. In *ICLR*, 2022. arXiv:2102.06571.

- [156] J. Sun, C. Gong, Y. Du, and Q. Liu. Beam search optimized batch Bayesian active learning. In *AAAI*, 2023.
- [157] J. T. Springenberg, A. Klein, S. Falkner, and F. Hutter. Bayesian optimization with robust Bayesian neural networks. In *NeurIPS*, 2016.
- [158] R. Rahaman and A. G. Thiery. Uncertainty quantification and deep ensembles. In *NeurIPS*, 2021. arXiv: 2007.08792.
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